

Analysis Matrix quiz 1 2025-26

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P5

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 6 & -2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 3 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}$$

For questions 1-6, reference the matrices above. Perform the given operations or state that it does not exist (DNE) [2 points each]

1. $4B - C$

$$\begin{bmatrix} 8 & 4 \\ -4 & 4 \end{bmatrix} - \begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix} =$$

$$\begin{bmatrix} 9 & 1 \\ -6 & 0 \end{bmatrix} \checkmark$$

2. D^T

$$\begin{bmatrix} 2 & -1 & 1 \\ 3 & 1 & -1 \end{bmatrix} \checkmark$$

3. BA $2 \times 2 \quad 2 \times 3$

$$\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 6 & -2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 0 & 2 \\ 4 & -3 & -1 \end{bmatrix} \checkmark$$

4. CD $2 \times 2 \cdot 3 \times 2$

DNE $\checkmark$

5. DA $3 \times 2 \quad 2 \times 3$

$$\begin{bmatrix} 2 & 3 \\ -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 6 & -2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 22 & -4 & 2 \\ 4 & -3 & -1 \\ -4 & 3 & 1 \end{bmatrix} \checkmark$$

$$\frac{2}{5} + \frac{3}{5} = 1$$

$$-\frac{3}{10} + \frac{3}{10} = 0$$

$$-\frac{4}{5} + \frac{4}{5} = 0$$

$$\frac{6}{10} + \frac{4}{10} = 1$$

6. Using the matrix C above, create 2 different matrices E (besides $E = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$) such that $CE=EC$.

C^{-1} & I

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$E = \begin{bmatrix} -2/5 & 3/10 \\ 1/5 & 1/10 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -2/5 & 3/10 \\ 1/5 & 1/10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix} = -4 - 6 = -10$$

$$[2] \quad \frac{1}{-10} \begin{bmatrix} 4 & -3 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} \frac{4}{10} & \frac{3}{10} \\ \frac{2}{10} & \frac{1}{10} \end{bmatrix}$$

7. Solve the system of equations using matrices. Clearly show all your steps. [4]

$$2x + y = 19$$

$$x - y = -1$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 19 \\ -1 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 19 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 19 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{19}{3} - \frac{1}{3} \\ \frac{19}{3} + \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{18}{3} \\ \frac{21}{3} \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

Final Answer: $x=6 \quad y=7 \quad \checkmark$

$$2(6) + 7 = 12 + 7 = 19 \checkmark$$

$$6 - 7 = -1 \checkmark$$

q1 + route:

$$XA = B \quad X \cdot A \cdot A^{-1} = B \cdot A^{-1}$$

$$X = BA^{-1} \quad X \cdot I = BA^{-1}$$

$$\begin{bmatrix} -1 & -\sqrt{3} \\ -\frac{2}{3} & -\frac{1}{6} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -7 & -13 \end{bmatrix} = \begin{bmatrix} -1 + \sqrt{3} \\ -\frac{2}{3} + \frac{1}{6} \end{bmatrix}$$

✓ match

8. Given the matrices $A = \begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$, solve for the matrix X in the equation $X(A - 5B) = C$.

Simplify your answer.

$$X \cdot \left(\begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 5 \end{bmatrix} \right) = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$$

$$X \cdot \begin{bmatrix} -13 & -1 \\ 7 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} -13 & -1 \\ 7 & 1 \end{bmatrix}^{-1}$$

$$X = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix} \frac{1}{-6} \begin{bmatrix} 1 & 1 \\ -7 & -13 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{6} & -\frac{1}{6} \\ \frac{7}{6} & \frac{13}{6} \end{bmatrix}$$

$$= \begin{bmatrix} -1 + \frac{7}{3} & -1 + \frac{13}{3} \\ -\frac{2}{3} + \frac{7}{6} & -\frac{4}{3} + \frac{13}{6} \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & \frac{10}{3} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

✓ match

$$-\frac{3}{3} + \frac{7}{3} = \frac{4}{3}$$

$$\frac{3}{3} + \frac{13}{3} = \frac{16}{3} \quad -\frac{4}{6} + \frac{7}{6} = \frac{3}{6} = \frac{1}{2}$$

$$\frac{4}{6} + \frac{13}{6} = \frac{17}{6}$$

$$-\frac{3}{3} + \frac{13}{3} = \frac{10}{3}$$

$$-\frac{4}{6} + \frac{13}{6} = \frac{9}{6} = \frac{3}{2}$$

9. Solve for both values of x that would make the determinant of matrix $F = 0$. [4]

$$F = \begin{bmatrix} x & 1 & 0 \\ 2 & x & 1 \\ 0 & 3 & 4 \end{bmatrix}$$

$4x^2$

$$4x^2 - (3x + 8) = 0$$

$$4x^2 - 3x - 8 = 0$$

$$X = \frac{3 \pm \sqrt{9 - 4(4)(-8)}}{8} = \frac{3 \pm \sqrt{137}}{8}$$

$$\frac{32}{4} \cdot 8$$

$$2 \cdot 16$$

$$9 - (16)(8) = 9 + 128 = 137$$

$$9 + 128 = \frac{4}{16} \cdot \frac{8}{128} + \frac{9}{137}$$

$$X = \frac{3 + \sqrt{137}}{8}$$

$$X = \frac{3 - \sqrt{137}}{8}$$

